

Questions

Anand Kale

August 23, 2026

Rough Work

Q1. Denoting the number of solutions of $f(x) \equiv k \pmod{m}$ by $N(k)$ prove that $\sum_{k=1}^m N(k) = m$

Solution: let ℓ_k be the set of congruence classes (solutions) satisfying $f(x) \equiv k \pmod{m}$ so, $\ell_k \subseteq \mathcal{C}_m$ where $\mathcal{C}_m$ is the complete residue system modulo m , more precisely

$$\ell_k = \{[x] \in \mathcal{C}_m : f(x) \equiv k \pmod{m}\}$$

where $[x]$ denotes the congruence class $x \pmod{m}$. So $N(k) = |\ell_k|$
Now think if we can say that ℓ_k for $1 \leq k \leq m$ are mutually exclusive and exhaustive, that is

$$\bigcup_{k=1}^m \ell_k = \mathcal{C}_m$$

and

$$\ell_i \cap \ell_j = \phi \quad \forall i \neq j$$

for claim of mutually exclusiveness, we can proceed as following- suppose some residue classes belong to both ℓ_i and ℓ_j then

$$\begin{aligned} f(x) &\equiv i \pmod{m} \\ f(x) &\equiv j \pmod{m} \\ \therefore i &\equiv j \pmod{m} \end{aligned}$$

but $i, j \in \{1, 2, \dots, m\}$ so they can't be equivalent unless $i = j$.
Thus for $i \neq j$

$$\ell_i \cap \ell_j = \phi \quad \forall i \neq j$$

for the second claim- Exhaustiveness, take any residue x modulo m , now $f(x)$ is an integer so it must be some integer modulo m which can be represented by any of $1, 2, \dots, m$ so there is some $k \in \{1, 2, \dots, m\}$ such that $f(x) \equiv k \pmod{m}$ hence $x \in \ell_k$.

Therefore every element in $\mathcal{C}_m$ belong to some ℓ_k , hence

$$\bigcup_{k=1}^m \ell_k = \mathcal{C}_m$$

the above claims are sufficient to state that

$$\begin{aligned} \sum_{k=1}^m N(k) &= \sum_{k=1}^m |\ell_k| \\ &= |\mathcal{C}_m| \\ &= m \end{aligned}$$